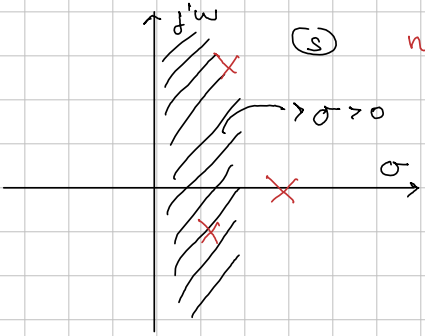
POTREBNI I DOVOLJNI UVJETI ZA PRF

Z & Y → Funkcije imitancija

LTI mreže

$$W(s) = \frac{s^4 + 3s^3 + 2s^2 + s + 1}{s^3 + 2s^2 + s + 2} = \frac{P(s)}{Q(s)}$$



nemoguće da su polovi desno

$W(s) \to \mathbb{R}$ — pozitivne realne fje

$W(s) > 0$, ako $\sigma > 0$

+ ako nije, onda to nije mičija imitancija

PRF (poz. real. fja)

UVJETI:

a) $W(s)$ je regularna na $\sigma > 0$ (nema polova desno)

b) Polovi na jω osi – jednostruki sa realnim i pozitivnim rezidumima (uključujući $s=0$, $s=\infty$)

↳ ako je isključivo imaginarna ⇒ jednostruki „X"

c) $\mathrm{Re}\{W(s)\}\Big|_{s=j\omega} \geq 0$ za sve ω osim u polovima

→ prije toga ustanoviti:

⇒ polinomi u brojniku i nazivniku $W(s)$ imaju Re koef.

⇒ stupnjevi polinoma u brojniku odnosno nazivniku $W(s)$ mogu se razlikovati najviše za $\underline{1}$

Primjer 1.) Je li funkcija $W(s)$ funkcija imitancije (P.R.F)?

$$W(s) = \frac{s^3+s^2+2s}{s^3+s^2+s+1} = \frac{s(s^2+s+2)}{(s+1)(s^2+1)} = \frac{P(s)}{Q(s)}$$

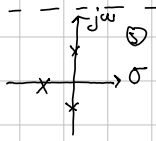
1) koeficijenti su Re i >0 ✓

2) stupanje se razlikuje za max 1 ✓

} intro uvjeti

3) polovi: $s_{p1} = -1$

$s_{p2,3} = \pm j$



} nisu desno ✓ jednostruki ✓

4) Polovi s_{p2} i s_{p3} (ovi na $j\omega$) su jednostruki sa rezidumima:

$$k_{p1} = k_{p2} = \left(\frac{s^2+1}{2s} W(s)\right)\Bigg|_{s^2=-1} = \left(\frac{\cancel{s^2+1}}{2\cancel{s}} \cdot \frac{\cancel{s}(s^2+s+2)}{(s+1)\cancel{(s^2+1)}}\right)\Bigg|_{s_p=\pm j} = \left(\frac{s^2+s+2}{2(s+1)}\right)\Bigg|_{s=\pm j} = \frac{1\pm j}{2(1\pm j)} = \frac{1}{2} > 0$$

✓ pozitivan ✓

5) $$\mathrm{Re}\{W(s)\}\Big|_{s=j\omega} \overset{*}{=} \frac{-j\omega^3-\omega^2+2j\omega}{-j\omega^3-\omega^2+j\omega+1} = \frac{(-\omega^2)+j(2\omega-\omega^3)}{(-\omega^2+1)+j(\omega-\omega^3)}$$

$$\mathrm{Re}\{W(s)\}\Big|_{s=j\omega} = \frac{\omega^2}{\omega^2-1} \geq 0 \quad \text{za sve } \omega \text{ osim u polovima}$$

DA ✓

Primjer 2.) Je li fja $W(s)$ fja imitancije (P.R.F)?

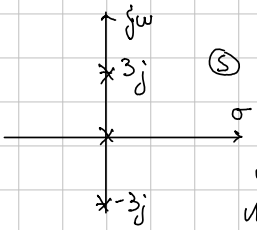
$$W(s) = \frac{(s^2+1)(s^2+2)}{s(s^2+3)} = \frac{s^4+3s^2+2}{s^3+3s}$$

→ koef. realni i poz ✓

→ stupnjevi razlikuju za max 1

a) Polovi: $s(s^2+3) = 0$

$s_{p1} = 0 \qquad s_{p2,3} = \pm 3j$



✓ nema desno ✓ jednostruki na jω

b) Reziduumi:

$$k_{p1} = \frac{s}{1} \cdot W(s)\Big|_{s=0} = \frac{(s^2+1)(s^2+2)}{(s^2+3)}\Bigg|_{s=0} = \lim_{s\to 0} \quad ✓$$

s

$$k_{p2} = \left(\frac{s^2+3}{2s} \cdot W(s)\right)\Bigg|_{s^2=-3} = \frac{(s^2+1)(s^2+2)}{2s^2}\Bigg|_{s^2=-3} = \frac{\overset{-2}{(-3+1)}\overset{-1}{(-3+2)}}{-6} = \frac{2}{-6} = \frac{-1}{3}$$

s^2+3

NE

Primjer 3.) Jesu li sljedeće fje PRF

3a) $$W(s) = \frac{s^3+6s^2+2s+1}{s+4} = \frac{P(s)}{Q(s)}$$

1) Re, >0 ✓ 2) ↯ stupnjevi se razlikuju za više od 1 NE

3b) $$W(s) = \frac{s^2+2s+1}{s^2} = \frac{P(s)}{Q(s)}$$ 1) ✓ 2) ✓

Polovi $\Rightarrow$ $s_{0_{1,2}} = 0 \rightarrow$ ↯ dvostruki NE

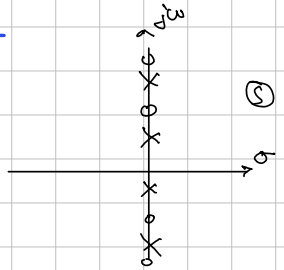
3c) $$W(s) = \frac{s^3+7s^2+15s+9}{s^4+6s^2+9}$$ 1) Re, >0 ✓ 2) Stupnjevi max ± 1 ✓

$\rightarrow$ Polovi: $(s^2+3)^2=0 \rightarrow s_{p_{1,2}}^2+3=0 \rightarrow s_{p_{1,2}} = \pm\sqrt{3}j$ $s_{p_{3,4}} = \pm\sqrt{3}j$ ↳ dvostruki! NE

Podjela imitancija

- Funkcija Impedancije $Z = R + j\cdot X$ (R: Rezistancija, X: Reaktancija)
- Funkcija Admitancije $Y = G + j\cdot B$ (G: kondukcija, B: susceptancija)

Sinteza LC dvopola po Fosteru



UVJETI:

- neparna racionalna funkcija od „s"
- ima jednostruke polove (samo na jω osi)
- reziduumi u tim polovima realni i pozitivni (ili polovi i nule alterniraju)

$$W(s) = k \cdot \frac{(s^2+w_1^2)\cdot(s^2+w_3^2)\cdots(s^2+w_{2n+1}^2)}{s\cdot(s^2+w_2^2)(s^2+w_4^2)\cdots(s^2+w_{2n}^2)} \rightarrow \text{fja neparna}$$

Fosterovi oblici

RASTAV općeg oblika fje imitancije

$$W(s) = k \cdot \frac{(s^2+w_1^2)\cdot(s^2+w_3^2)\cdots(s^2+w_{2n+1}^2)}{s\cdot(s^2+w_2^2)(s^2+w_4^2)\cdots(s^2+w_{2n}^2)}$$

zapamtiti!

rastav na parc. razlomke

Na parcijalne razlomke:

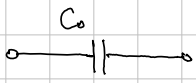
$$W(s) = \frac{k_0}{s} + \sum_i \frac{2k_i\cdot s}{s^2+w_i^2} + k_\infty \cdot s$$

PRI ČEMU SU

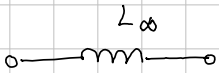
$$k_0 = \lim_{s\to 0}(s\cdot W(s)) \qquad k_i = \lim_{s^2\to -w_i^2}\left(\frac{s^2+w_i^2}{2s}\cdot W(s)\right) \qquad k_\infty = \lim_{s\to\infty}\left(\frac{1}{s}\cdot W(s)\right)$$

I. Fosterov oblik (kanonski)

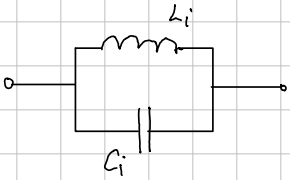
→ rastavljamo impedanciju $W(s) = Z(s)$



$$Z_C(s) = \frac{1}{sC_0} = \frac{k_0}{s} = \frac{1}{s\cdot\frac{1}{k_0}} \qquad C_0 = \frac{1}{k_0}$$



$$Z_L(s) = sL_\infty = s\cdot k_\infty \qquad L_\infty = k_\infty$$

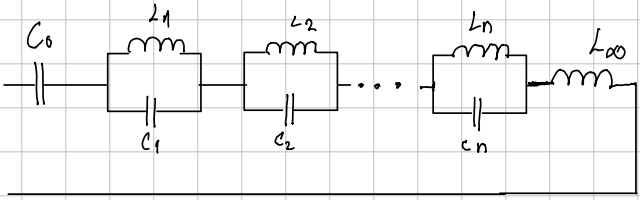


$$Z_i = \frac{\frac{1}{sC_i}\cdot sL_i}{sL_i + \frac{1}{sC_i}} = \frac{\frac{L_i}{C_i}}{\frac{s^2L_iC_i+1}{sC_i}} = \frac{sL_i}{s^2L_iC_i+1}$$

$$Z_i = \frac{1}{C_i}\,\frac{s}{s^2+\frac{1}{L_iC_i}} = \frac{2k_i\cdot s}{s^2+w_i^2} \rightarrow C_i = \frac{1}{2k_i} \qquad L_i = \frac{2k_i}{w_i^2}$$

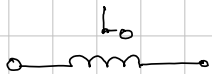
Realizacija

$Z(s) \rightarrow$



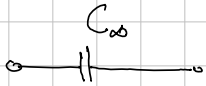
$$Z(s) = \frac{k_0}{s} + \sum_i \frac{2k_i\cdot s}{s^2+w_i^2} + k_\infty\cdot s$$

▶ II. Fosterov oblik (kanonski) $W(s) = Y(s)$



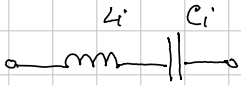
$$Y_L(s) = \frac{1}{sL_0} = \frac{k_0}{s} = \frac{1}{s \cdot \frac{1}{k_0}}$$

$$L_0 = \frac{1}{k_0}$$



$$Y_C(s) = sC_\infty$$

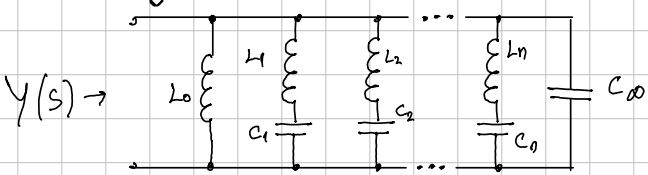
$$C_\infty = k_\infty$$



$$Y_i(s) = \frac{1}{sL_i + \frac{1}{sC_i}} = \frac{1}{L_i} \cdot \frac{s}{s^2 + \frac{1}{L_i C_i}} = \frac{2k_i \cdot s}{s^2 + W_i^2}$$

$$L_i = \frac{1}{2k_i} \qquad C_i = \frac{2k_i}{W_i^2}$$

Realizacija:



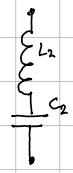
$$Y(s) = \frac{k_0}{s} + \sum_i \frac{2k_i \cdot s}{s^2 + W_i^2} + k_\infty \cdot s$$

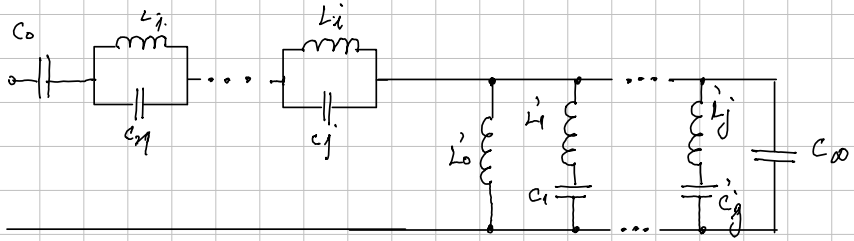
Moguća komb. prvog i drugog Fosterovog oblika

- pojedine dijelove $W(s)$ raspisujemo kao

$$W(s) = \frac{k_0}{s} + \sum_i \frac{2k_i \cdot s}{s^2 + W_i^2} + W_1(s) \qquad (i < n)$$

$$W'(s) = \frac{1}{W_1(s)} = \frac{k_0'}{s} + \sum_j \frac{2k_j' \cdot s}{s^2 + W_j'^2} + k_\infty' \cdot s$$





PRIMJERI - FOSTER

Primjer 1: Zadani rasporedi unutarnjih polova i nula realt. dvopola prema slici.

Odrediti Fosterove kanonske oblike ako je zadano:

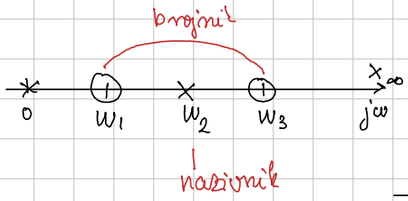
$\omega_1 = 2$

$\omega_2 = 4$

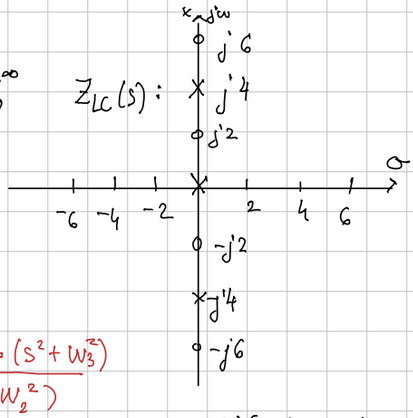
$\omega_3 = 6$

Za $\omega_a = 16$

$\hookrightarrow jX(\omega_a) = j3$



$Z_{LC}(s)$:



Rješenje: fja imp: $Z(s) = H \cdot \dfrac{(s^2+\omega_1^2)\cdot(s^2+\omega_3^2)}{s\cdot(s^2+\omega_2^2)}$

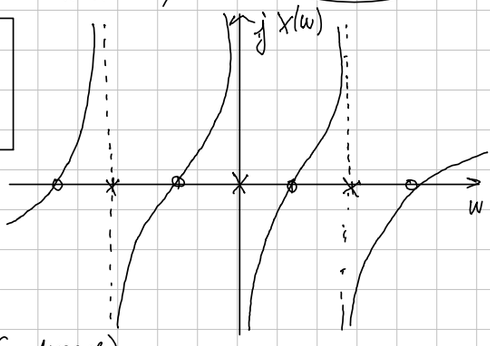
$s = j\omega \rightarrow$

$$jX(\omega) = H \cdot \frac{(\omega^2-\omega_1^2)(\omega^2-\omega_3^2)}{-j\omega\,(\omega^2-\omega_2^2)} = jH\,\frac{(\omega^2-\omega_1^2)(\omega^2-\omega_3^2)}{\omega(\omega^2-\omega_2^2)}$$

$$jX(\omega_a) = jH\,\frac{(16^2-2^2)(16^2-6^2)}{16(16^2-4^2)} = j3$$

$$H\cdot\frac{(256-4)(256-36)}{16(256-16)} = 3 \rightarrow H = 3\cdot\frac{16\cdot 240}{252\cdot 220} \rightarrow H = 0.208$$

$$\Rightarrow \boxed{Z(s) = 0.208\,\frac{(s^2+4)(s^2+36)}{s(s^2+16)}}$$



PRVI FOSTEROV OBLIK (LC dvopol)

$$Z(s) = 0{,}208\,\frac{(s^2+4)(s^2+36)}{s(s^2+16)} = \frac{k_0}{s} + \frac{2k_2\cdot s}{s^2+16} + k_\infty\cdot s$$

Reziduumi

$$k_0 = \lim_{s\to 0}\left(s\cdot Z(s)\right) = 1.872$$

$$k_2 = \lim_{s\to -16}\left[(s+16)\cdot Z(s)\right] = 1{,}56$$

$$k_\infty = \lim_{s\to\infty}\left(Z(s)\right) = 0.208$$

$$C_0 = \frac{1}{k_0} = \frac{1}{1.8722} = \underline{\underline{0{,}535}}$$

$$L_\infty = k_\infty = \underline{\underline{0.208}}$$

$$C_2 = \frac{1}{2k_4} = \underline{\underline{0{,}321}} \qquad L_2 = \frac{2k}{\omega_i^2} = \underline{\underline{0.195}}$$

DRUGI FOSTEROV OBLIK (LC dvopol)

$$Y(s) = \frac{1}{0.208}\cdot\frac{s(s^2+16)}{(s^2+4)(s^2+36)} = \frac{k_0}{s} + \frac{2k_1\cdot s}{s^2+4} + \frac{2k_3\cdot s}{s^2+36} + k_\infty\cdot s$$

$$k_0 = \lim_{s\to 0}\left(s\cdot\frac{s(s^2+16)}{(s^2+4)(s^2+36)}\cdot\frac{1}{0.208}\right) = 0$$

$$k_1 = \lim_{s^2\to -\omega_1^2}\left[\frac{s^2+\omega_1^2}{2s}\cdot Y(s)\right] = 0{,}901 \qquad (\omega_1^2 = 4)$$

$$k_2 = \lim_{s^2\to -\omega_3^2}\left[\frac{s^2+\omega_3^2}{2s}\cdot Y(s)\right] = 1{,}502 \qquad (\omega_3^2 = 36)$$

$$k_\infty = \lim_{s\to\infty}\left[\frac{1}{s}\cdot Y(s)\right] = \lim_{s\to\infty}\frac{1}{0.208}\cdot\frac{(s^2+16)}{(s^2+4)(s^2+36)} = 0$$

$L_0 = \infty \qquad C_\infty = 0$

$L_1 = \dfrac{1}{2k_1} = 0{,}555$

$C_1 = \dfrac{2k_1}{\omega_1^2} = 0{,}451$

$L_3 = \dfrac{1}{2k_3} = 0{,}333$

$C_3 = \dfrac{2k_3}{\omega_3^2} = 0.083$

SINTEZA RC po Fosteru

RF $Z_{RC}(s)$ – fja impedancije RC dvopola

1) nule i polovi $\times 1$, <0 Re ili $\emptyset$

2) polovi i nule alterniraju na nepozitivnom dijelu realne osi

3) Kritična frekv. s najnižim modulom = pol

4) Multiplikativna konst $k_0 > 0$

▸ I. FOSTEROV oblik za RC

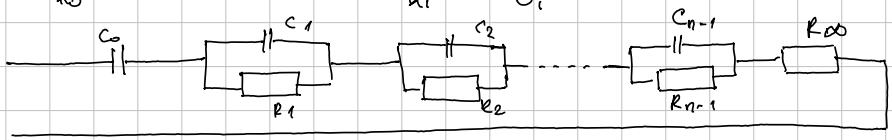
$$\Rightarrow Z_{RC}(s) = \frac{k_0}{s} + \sum_i \frac{k_i}{s+\sigma_i} + k_\infty$$

$$k_0 = \lim_{s\to 0}\left(s \cdot Z_{RC}(s)\right)$$

$$k_i = \lim_{s\to -\sigma_i}\left[(s+\sigma_i)\cdot Z_{RC}(s)\right]$$

$$k_\infty = \lim_{s\to\infty}\left(Z_{RC}(s)\right)$$

$$C_0 = \frac{1}{k_0} \qquad R_\infty = k_\infty \qquad C_i = \frac{1}{k_i},\ R_i = \frac{k_i}{\sigma_i}$$



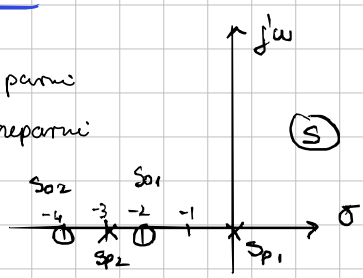
II. Fosterov oblik za RC

$$\Rightarrow Y_{RC}(s) = k_0 + \sum_i \frac{k_i \cdot s}{s+\sigma_i} + k_\infty \cdot s$$

bla bla ostalo u formulama

PRIMJERI - FOSTER

Primjer 1.) $Z_{RC}(s) = \dfrac{(s+2)(s+4)}{s(s+3)}$ → parni / → neparni



PRVI FOSTER:

$$Z_{RC}(s) = \frac{k_0}{s} + \sum_i \frac{k_i}{s+\sigma_i} + k_\infty$$

$$k_0 = \lim_{s\to 0}\left(s \cdot Z_{RC}(s)\right) = \lim_{s\to 0}\frac{(s+2)(s+4)}{s+3} = \lim_{s\to 0}\frac{s^2+6s+8 \,/\!:s^2}{s+3 \,/\!:s^2} = \frac{8}{3} = k_0$$

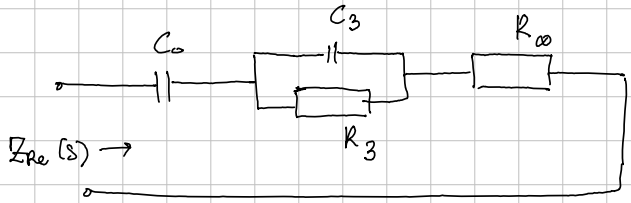
$$k_3 = \lim_{s\to -3}\left[(s+3)\cdot Z_{RC}(s)\right] = \lim_{s\to -3}\frac{s^2+6s+8}{s} = \left(\frac{9-18+8}{-3}\right) = \frac{1}{3} = 0.333$$

$$k_\infty = \lim_{s\to\infty}\frac{s^2+6s+8}{s+3} = \frac{1}{0} \Rightarrow k_\infty = \infty$$

* oni su ku nešto zezali
→ 1 se dobije

$$^*Z_{RC}(s) = 1 + \frac{8}{3s} + \frac{0.33}{s+3}$$

$C_0 = \dfrac{1}{k_0}$ → $\boxed{C_0 = \frac{3}{8}}$ $\quad$ $^*k_\infty = 1$ $\quad$ $\boxed{C_3 = 3}$ $\quad$ $\boxed{R_3 = \frac{1}{9}}$



DRUGI FOSTER:

$$Z_{RC}(s) = \frac{1}{Y(s)} \Rightarrow Y_{RC}(s) = \frac{s(s+3)}{(s+2)(s+4)}$$

$$Y_{RC}(s) = k_0 + \sum_i \frac{k_i \cdot s}{s+\sigma_i} + k_\infty \cdot s$$

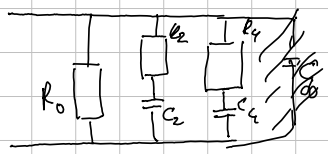
$$k_0 = \lim_{s\to 0}\left(\frac{\cancel{s}\cdot Y_{RC}(s)}{\cancel{s}}\right) = \lim_{s\to 0}\frac{s(s+3)}{(s+2)(s+4)} = 0$$

$$k_2 = \lim_{s\to -2}\frac{(s+3)}{s+4} = \frac{1}{2}$$

$$k_\infty = \lim_{s\to\infty}\left(\frac{Y_{RC}}{s}\right) = \lim_{s\to\infty}\frac{s+3}{(s+2)(s+4)}$$

$$k_4 = \lim_{s\to -4}\frac{(s+3)}{(s+2)} = \frac{1}{2}$$

$$k_\infty = 0$$

$\boxed{R_0 = \infty}$ $\quad$ $\boxed{C_\infty = 0}$ $\quad$ $\boxed{R_2 = 2}$ $\quad$ $\boxed{R_4 = 2}$

$\boxed{C_2 = 1/4}$ $\quad$ $\boxed{C_4 = \frac{1}{8}}$